

CB-SAFE: Code-Based Post-Quantum Secure Aggregation for Byzantine-Robust Federated Learning

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Abstract—Secure aggregation protects federated learning (FL) updates by revealing only their sum, but its key establishment breaks under a quantum adversary, and post-quantum FL today derives confidentiality almost exclusively from lattice assumptions, a single point of cryptanalytic failure. We present CB-SAFE, to our knowledge the first FL secure-aggregation framework whose post-quantum confidentiality is established by a code-based KEM (HQC, selected by NIST in March 2025 to diversify away from lattices) rather than a lattice KEM. CB-SAFE treats the KEM as a pluggable module: the same double-masking protocol runs over HQC or ML-KEM, and per-round mask seeds are derived from long-lived encapsulated secrets, so the KEM affects only a one-time setup. Measured on a 291 498-parameter model with 30 clients, per-round traffic is KEM-independent (2277 KiB per client), and the entire code-based premium is a one-time 15.2 vs. 3.8 KiB per client at NIST level 1 (49.4 vs. 7.7 KiB at level 5), under 0.03% of the traffic of a single training round. CB-SAFE further reconciles secure aggregation with Byzantine robustness by aggregating securely *within* clusters of c clients and applying robust rules (trimmed mean, median, Multi-Krum) *across* the revealed cluster sums, and we quantify the resulting privacy–robustness tension: the anonymity set grows with c while the probability a cluster sum stays uncontaminated decays as $(1 - f)^c$ under malicious fraction f . We prove a simulation-based privacy theorem, identify a *laundering* effect by which cluster averaging turns amplified sign-flip poisoning into plausible-magnitude inverted sums that defeat all coordinate-wise robust rules, and introduce CB-SAFE+: re-randomized clustering with root-anchored direction flags accumulates per-client suspicion across rounds (temporal group testing over private sums) and excludes attackers the server never observes individually, at zero additional leakage. Experiments across three datasets and two modalities (non-IID CIFAR-10 and FashionMNIST, and tabular Edge-IIoTset IoT intrusion detection) with label-flipping, sign-flipping, and backdoor attacks validate the framework: the laundering effect and CB-SAFE+’s recovery replicate on every dataset (CB-SAFE+ beats every baseline under sign-flip, Holm-corrected $p < 0.01$), and the implementation is released open-source.

Index Terms—Post-quantum cryptography, federated learning, secure aggregation, HQC, code-based cryptography, Byzantine robustness, cryptographic diversity.

I. INTRODUCTION

Federated learning (FL) trains a shared model across many clients without centralizing data, but the transmitted model updates leak private information: gradient-inversion and membership-inference attacks reconstruct training samples from individual updates. Secure aggregation closes this channel by ensuring the server observes only the *sum* of updates,

using pairwise additive masks that cancel on aggregation [1], [12]. The confidentiality of those masks rests on key establishment that Shor’s algorithm breaks once a cryptographically relevant quantum computer exists, and traffic recorded today can be decrypted then (“harvest-now, decrypt-later”).

The standard response replaces classical key exchange with a post-quantum KEM, and nearly all post-quantum FL does so with *lattice* cryptography [4]–[7], [17]. Lattices are efficient and standardized (ML-KEM, FIPS 203 [3]), but concentrating an entire ecosystem’s confidentiality on one hardness assumption is a systemic risk: a single cryptanalytic advance would break every such deployment simultaneously. NIST recognized exactly this risk when, in March 2025, it *selected* the code-based KEM HQC (whose security reduces to syndrome decoding of quasi-cyclic codes, not to lattice problems) as a diversified backup to ML-KEM [2]. That diversity has not reached federated learning.

A second, orthogonal tension is that secure aggregation and Byzantine robustness pull in opposite directions: hiding individual updates denies the server exactly the visibility that poisoning defenses need. Recent work combines robustness with *lattice*-based post-quantum aggregation [7], or achieves cluster-based robust aggregation on *quantum hardware* [8]; no prior framework provides both under a code-based, classical post-quantum assumption.

This paper presents CB-SAFE (Code-Based Secure Aggregation for Federated Learning), making four contributions:

- 1) **A KEM-agnostic secure-aggregation protocol** (Definition 1) that formalizes Bonawitz-style double masking over any IND-CCA2 KEM, with per-round mask seeds derived from long-lived encapsulated secrets so key encapsulation is a one-time cost. The post-quantum primitive becomes a single replaceable module.
- 2) **The first code-based instantiation, scope-qualified:** to our knowledge, CB-SAFE is the first FL secure-aggregation framework whose post-quantum confidentiality comes from the NIST-selected code-based KEM (HQC) rather than a lattice KEM, at matched NIST security levels.
- 3) **A quantified reconciliation of privacy and robustness:** secure aggregation *within* clusters of c clients, robust aggregation (trimmed mean, median, Multi-Krum) *across* cluster sums, and an explicit trade-off law (the

anonymity set is c while the clean-cluster probability decays as $(1-f)^c$ validated experimentally under three attacks. We further identify and analyze a *laundering* effect: cluster averaging turns amplified sign-flip poisoning into plausible-magnitude, direction-inverted cluster sums that defeat every coordinate-wise robust rule (Proposition 1).

- 4) **CB-SAFE+, a defense that identifies hidden attackers:** per-round re-randomized clustering plus direction flags against a small server root dataset accumulate per-client suspicion (temporal group testing over privacy-preserving observations) and exclude malicious clients the server has never observed individually. The flags are post-processing of already-revealed cluster sums, so the defense adds *no* leakage beyond CB-SAFE itself (Theorem 1).
- 5) **An honest, measured cost analysis:** on a 291k-parameter model, per-round traffic is KEM-independent and the full code-based premium is a one-time 15.2 KiB vs. 3.8 KiB per client (NIST level 1), i.e., under 0.03% of one round’s traffic; HQC’s $\sim 200\times$ slower operations contribute about 2 s (one-time, 30 clients) versus 0.02 s for ML-KEM.

II. RELATED WORK

Post-quantum FL is lattice-dominated. PQSF reconstructs masks with lattice multi-stage secret sharing [4]; Qin and Xu build single-round aggregation from a lattice key-homomorphic PRF [5]; Narsimhulu *et al.* aggregate Dilithium-style lattice signatures [6]; and a 2026 framework combines reputation-weighted Byzantine-robust aggregation with Kyber/lattice-based secure aggregation for IoT threat intelligence [7]. Surveys confirm lattice dominance across post-quantum deployment [17]. Non-lattice exceptions are rare and code-free: multivariate cryptography with blockchain [9], and CQSA, which partitions clients into clusters aggregated via GHZ states on near-term *quantum hardware* [8]. CB-SAFE differs from all of the above on the confidentiality assumption: syndrome decoding of quasi-cyclic codes, via a standard-track classical KEM [2], with the KEM exposed behind an agnostic interface so lattice and code-based variants differ in exactly one module.

Cluster-based robust aggregation. Outside the post-quantum setting, reconciling Byzantine resilience with secure aggregation is an established goal: BREA does so through secret-shared distance computation [16], and CQSA through clustered aggregation on quantum hardware [8]. Aggregating within groups and inspecting group sums is thus a known structure; our contribution is not the cluster topology itself but (i) its first classical post-quantum, code-based instantiation and (ii) an explicit, experimentally validated statement of the privacy–robustness trade-off it induces (Section III-C).

III. THE CB-SAFE FRAMEWORK

A. System and Threat Model

A server coordinates N clients running FedAvg over non-IID data partitions. Clients are partitioned into $k = N/c$ fixed clusters of size c . The server is honest-but-curious; a fraction f of clients are malicious and submit poisoned updates; clients may drop out mid-round. Within a cluster, mask recovery uses a t -of- c threshold with $t = \lfloor c/2 \rfloor + 1$, so privacy holds against the server colluding with up to $t - 1$ members of a cluster, and any $c - t$ members can drop without stalling the round. Update authentication is out of the code-based scope: HQC cannot sign and no code-based signature is standardized, so where signatures are required we compose the lattice-based ML-DSA and state explicitly that only the confidentiality layer is code-based.

B. KEM-Agnostic Masked Aggregation

Definition 1 (KEM-agnostic cluster masking): Let $\mathcal{K} = (\text{KeyGen}, \text{Encaps}, \text{Decaps})$ be an IND-CCA2 KEM, H a key-derivation function, and G a PRG into $(\mathbb{Z}_{2^{64}})^d$. At setup, each client i publishes pk_i ; for each cluster pair $i < j$, client i computes $(ct_{ij}, ss_{ij}) = \text{Encaps}(pk_j)$ and j recovers $ss_{ij} = \text{Decaps}(sk_j, ct_{ij})$. In round r , client i quantizes its update x_i to $q(x_i) \in (\mathbb{Z}_{2^{64}})^d$, draws a self-mask seed b_i^r , Shamir-shares it t -of- c within its cluster, and submits

$$y_i = q(x_i) + G(b_i^r) + \sum_{j>i} G(H(ss_{ij}, r)) - \sum_{j<i} G(H(ss_{ij}, r)),$$

all mod 2^{64} , sums over the cluster peers of i . The server adds each cluster’s submissions (pairwise masks cancel), reconstructs each survivor’s b_i^r from t shares and removes $G(b_i^r)$; for a dropped client d , each survivor j reveals the per-round seed $H(ss_{dj}, r)$ and the server removes the orphaned mask halves. The server thus obtains exactly $\sum_{i \in \text{alive}} q(x_i)$ per cluster.

Correctness is exact: masking is modular-arithmetic exact, and the only deviation from plain float aggregation is fixed-point quantization (2^{-16} per coordinate). Because encapsulation happens once and per-round seeds come from $H(ss_{ij}, r)$, per-round messages contain no KEM material at all: the masked update is $8d$ bytes regardless of the KEM, and the KEM’s key and ciphertext sizes appear only in the one-time setup. Instantiating $\mathcal{K}$ with HQC gives the code-based variant; with ML-KEM [3], the lattice baseline.

Theorem 1 (Privacy against honest-but-curious coalitions): Assume $\mathcal{K}$ is IND-CCA2, H is a secure KDF, and G is a secure PRG. Fix a cluster C of size c with threshold $t = \lfloor c/2 \rfloor + 1$, and let the adversary $\mathcal{A}$ comprise the honest-but-curious server and any $T \subset C$ with $|T| \leq t - 1$. There exists a PPT simulator that, given only the corrupted clients’ inputs and the sum $\sum_{i \in \text{alive}(C) \setminus T} q(x_i)$, produces a view computationally indistinguishable from $\mathcal{A}$ ’s view of a CB-SAFE round. The CB-SAFE+ flag bits are a deterministic function of the revealed cluster sums and server-local data, so CB-SAFE+ leaks nothing beyond CB-SAFE.

Proof sketch. A hybrid argument. (H1) Replace each pairwise secret ss_{ij} between honest clients by a uniform string:

distinguishing H1 from the real game breaks IND-CCA2 of $\mathcal{K}$. (H2) Replace the per-round seeds $H(ss_{ij}, r)$ and mask streams $G(\cdot)$ by uniform: breaks the KDF/PRG. In H2 every honest submission y_i is uniform subject to the alive-honest submissions summing to the revealed cluster sum, which the simulator can sample directly. Shamir shares held by $\leq t-1$ corrupted parties are uniform by perfect secrecy below the threshold, and a dropped client's revealed seeds $H(ss_{dj}, r)$ are independent of other rounds' seeds under the KDF assumption, so both recovery paths are simulatable. Distinct rounds use distinct seeds, so views compose across rounds. $\square$

The anonymity set of an honest client is therefore its cluster's alive honest-member count, and re-randomizing the partition each round (CB-SAFE+) does not change the per-round argument because each round reveals sums of fresh, independently masked updates.

C. Robust Aggregation Across Cluster Sums

The server sees k cluster sums $S_1, \dots, S_k$ (means after dividing by alive counts) and no individual update. CB-SAFE applies a robust rule across the cluster means: coordinate-wise trimmed mean (trim β per side) or median [14], or Multi-Krum [13]. This resolves the hide-versus-inspect tension structurally (privacy is enforced below the cluster boundary, robustness operates above it) at a quantifiable price. A cluster mean is contaminated if it contains at least one malicious member, so with malicious fraction f and random cluster assignment,

$$\Pr[\text{cluster clean}] = (1 - f)^c,$$

and the expected number of clean cluster means is $k(1 - f)^c$. Larger c improves privacy (bigger anonymity set) but shrinks the clean fraction the robust rule can rely on; a trimmed mean discarding β per side tolerates at most $\lfloor \beta k \rfloor$ contaminated clusters. c is therefore a tunable privacy-robustness dial, and Section V traces its consequences empirically. Setting $c=1$ recovers classical (privacy-free) robust aggregation; setting $c=N$ recovers full secure aggregation with no robustness.

The contamination probability, however, understates the problem: cluster averaging also *disguises* what it contaminates.

Proposition 1 (Laundering by cluster averaging): Let each honest update be h (up to noise) and let m members of a cluster of size c submit the scaled sign-flip $-\gamma h$. The cluster mean is $\mu = \frac{c-m(1+\gamma)}{c} h$. Its magnitude equals an honest cluster's ($|\mu| = |h|$) exactly when $\gamma = 2c/m - 1$, while its direction is inverted. At the common attack setting $\gamma=5$ with $c=3$, $m=1$ this holds with equality: the poisoned cluster sum is indistinguishable from honest by any magnitude- or order-statistic test, and a near-symmetric $\pm h$ population drives coordinate-wise trimmed mean and median toward 0, stalling learning entirely.

This laundering effect explains the empirical collapse of every coordinate-wise rule in Section V and motivates direction-based detection: what cluster averaging cannot hide is the *sign* of the update.

D. CB-SAFE+: Temporal Group Testing over Cluster Sums

CB-SAFE+ turns the laundering signature into an identification mechanism, using three ingredients. (i) *Re-randomized clustering*: the partition is redrawn every round (pairwise secrets with all peers are established once at setup, so re-clustering adds no per-round KEM cost). (ii) *Loss-probe flags*: the server maintains a small root dataset ($|D_0|=200$ samples held out from all clients, the standard trust-bootstrapping assumption [10], [11]), applies each cluster mean to the current model, and flags every cluster whose mean *increases* root loss. Flags are computed from already-revealed cluster sums and server-local data, so they add no leakage (Theorem 1). The probe is the product of two measured failures. Anchor-free direction flags are sign-ambiguous: the poisoned cluster sums form their own coherent cloud at $-h$, and a Krum-based reference locks onto that cloud in a large fraction of rounds. Anchored *cosine* flags break the symmetry but lose specificity under non-IID data: an $m=1$ poisoned cluster mean is dominated by $-\gamma$ times one client's idiosyncratic direction, nearly orthogonal to the balanced anchor, so its sign test approaches a coin flip (at $f=0.2$ we measured cosine flags hitting clean clusters as often as dirty ones, and suspicion never separated). The loss probe is immune to this decorrelation because a γ -amplified ascent direction raises root loss regardless of whose data dominates the cluster sum. (iii) *Suspicion accumulation and exclusion*: every member of a flagged cluster gains one suspicion count. A malicious client sits in a flagged cluster in essentially every round, while an honest client is flagged only when randomly co-clustered with an attacker, i.e., with probability $p_h = 1 - (1 - f)^{c-1} < 1$. After R rounds the flagged fractions separate (by Hoeffding, misclassification probability decays as $e^{-2R(\tau-p_h)^2}$ for a threshold τ between p_h and 1), and clients whose fraction exceeds τ (0.85 here, after a 5-round warmup) are excluded. The server thereby identifies individual attackers it has never observed individually: group testing where the pools are the privacy-preserving cluster sums and the tests repeat over training rounds.

IV. IMPLEMENTATION

CB-SAFE is implemented in Python on liboqs 0.15.0 (via `liboqs-python`) and PyTorch, with the KEM selected by a one-line configuration change among HQC-128/192/256 and ML-KEM-512/768/1024 (Kyber round-3 parameter sets are retained for comparison with older literature). Masks are ChaCha20 keystreams; per-round seeds use HKDF-SHA256; self-mask seeds are shared with Shamir's scheme over $\text{GF}(2^{127} - 1)$. *HQC provenance*: liboqs 0.15.0 ships the PQClean constant-time implementation of the Round-4 HQC specification (2023-04-30), the version NIST evaluated and selected [2]; the finalized standard's parameters may differ slightly once published (a draft is expected as FIPS 207). Federated training is simulated in-process; every protocol byte count and crypto timing reported below is measured from real objects on the wire path of the protocol (public keys, ciphertexts, shares, masked vectors), not modeled. *Reproducibility*: Python 3.12, PyTorch 2.5.0 (CUDA 12.4), liboqs-

python 0.15.0, on a single NVIDIA RTX 2060 (6 GB); one training round of 30 clients takes ≈ 21 s. All runs use fixed seed 0 with deterministic partitioning and cluster assignment; CIFAR-10 and FashionMNIST results are averaged over three seeds (Edge-IIoTset over one), significance is by paired t -test with Holm correction, and every experiment is reproducible from a per-run CSV and the released scripts.

V. EVALUATION

A. Setup

Task and data. Primary task: CIFAR-10 partitioned across $N=30$ clients by a Dirichlet($\alpha=0.5$) label-skew split; a CNN with $d=291\,498$ parameters; one local epoch per round (SGD, lr 0.01, momentum 0.9, batch 64); $k=10$ clusters of $c=3$ unless stated. Generalization (Section V-E) adds Fashion-MNIST (same CNN) and Edge-IIoTset (a 15-class tabular IoT intrusion-detection set with a multilayer perceptron), each Dirichlet-partitioned identically. **Configurations.** Plain FedAvg (no cryptography); CB-SAFE/lattice (ML-KEM); CB-SAFE/code-based (HQC), at matched NIST levels. **Attacks.** Label flipping ($y \rightarrow 9 - y$), sign flipping (-5Δ), and a backdoor [15] (3×3 trigger patch, target class 0, 50% local poison rate) at malicious fractions $f \in \{0.1, 0.2, 0.3\}$, against aggregation rules mean, trimmed mean, median, Multi-Krum, and the CB-SAFE+ reputation defense (Section III-D; server root set of 200 samples held out from all clients). **Metrics.** Test accuracy; backdoor attack success rate (ASR: fraction of triggered non-target test samples classified as the target); communication bytes per client (setup vs. per-round); crypto wall-time; dropout-recovery cost.

B. Overhead: the Code-Based Premium Is a One-Time Cost

Table I reports measured per-client traffic and crypto time. Per-round traffic is identical across all six KEMs (2 277 KiB per client, dominated by the $8d$ -byte masked update plus 102 B of Shamir material) because Definition 1 keeps KEM objects out of the per-round path entirely. The KEM appears only in setup: at NIST level 1, HQC costs 15.2 KiB per client against ML-KEM’s 3.8 KiB ($4.0\times$; $6.4\times$ at level 5), and 2.0 s of one-time compute for all 30 clients against 0.02 s. Amortized over 40 training rounds (~ 89 MiB of per-client update traffic), the code-based premium is below 0.03% of total communication (Fig. 1). A round with 10% dropouts adds only seed-reveal messages (32 B per survivor-dropout pair) and kept server unmasking below 0.28 s in all configurations. The practicality question that motivated this work has a quantitative answer: *cryptographic diversity in FL secure aggregation is essentially free at the protocol level once key establishment is amortized*: the trade-off is a one-time setup constant, not a per-round tax.

C. Utility and Cryptographic Equivalence

Plain FedAvg reaches 59.0% test accuracy (mean of the final five of 40 rounds; 59.6% at round 40) on the non-IID split (Fig. 2). In every round, the same client updates were pushed through the full cryptographic pipeline twice

TABLE I
MEASURED PER-CLIENT OVERHEAD ($N=30$, $c=3$, $d=291\,498$). SETUP IS ONE-TIME; PER-ROUND TRAFFIC IS KEM-INDEPENDENT.

KEM	Setup (KiB)	Setup wall (s)	Round (KiB)	Mask (ms)
ML-KEM-512	3.8	0.02	2 277	25
HQC-128	15.2	2.0	2 277	26
ML-KEM-768	5.6	0.02	2 277	27
HQC-192	30.8	5.7	2 277	25
ML-KEM-1024	7.7	0.02	2 277	25
HQC-256	49.4	10.6	2 277	26

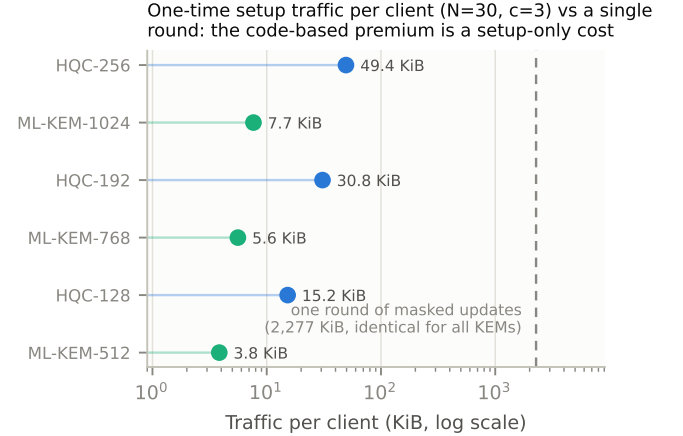


Fig. 1. One-time setup traffic per client versus a single round of masked updates (log scale). Even HQC-256’s setup is $\sim 2\%$ of one round’s traffic; per-round traffic is identical for all KEMs.

(once under HQC-128, once under ML-KEM-768), and every per-cluster secure sum was compared coordinate-wise against the plain sum. The maximum deviation across all 40 rounds and both KEMs was 2.28×10^{-5} , which *equals* the worst-case fixed-point rounding bound for a 3-element sum ($3 \cdot 2^{-17} = 2.29 \times 10^{-5}$): the masking arithmetic itself is exact, so CB-SAFE’s accuracy curve is identical to plain FedAvg by construction, not approximately. Both dropout modes were exercised: in round 10, two of three dropped clients landed in the same cluster, driving it below its $t=2$ threshold: the protocol excluded that cluster and recovered the remaining nine exactly; in round 20, three dropouts in three distinct clusters were all recovered by seed-reveal plus Shamir reconstruction. Measured in-run, setup cost 2.6 s (HQC-128) vs. 0.02 s (ML-KEM-768) for all 30 clients, and steady-state rounds took ~ 30 ms of client-side masking and ~ 0.29 s of server-side unmasking, negligible against the ~ 21 s of local training per round.

D. Robustness Under Attack

The contamination law holds. Across all runs, the realized fraction of clean cluster sums tracked $(1-f)^c$ closely: at $c=3$, observed 0.70/0.50/0.40 versus predicted 0.73/0.51/0.34 for $f=0.1/0.2/0.3$.

Sign-flip: laundering collapses coordinate-wise rules exactly at the predicted boundary (Table II, Fig. 3). At

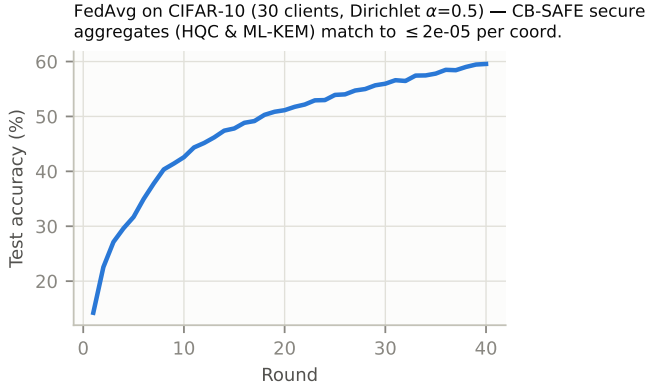


Fig. 2. Test accuracy of FedAvg on non-IID CIFAR-10; CB-SAFE’s secure aggregates (HQC and ML-KEM) are numerically equivalent to plain aggregation, so the accuracy curve is shared by all three configurations.

TABLE II
ROBUSTNESS OF AGGREGATION RULES OVER CLUSTER SUMS ($c=3$, $k=10$, SEED 0): FINAL TEST ACCURACY UNDER SIGN-FLIP AND ATTACK SUCCESS RATE (ASR, LOWER IS BETTER) UNDER THE BACKDOOR, AT MALICIOUS FRACTION f . BEST PER COLUMN IN BOLD. CB-SAFE+ ADDITIONALLY REPORTS ZERO HONEST FALSE-POSITIVE EXCLUSIONS IN EVERY CONFIGURATION.

Rule	Sign-flip acc. (%)			Backdoor ASR (%)		
	$f=.1$	$f=.2$	$f=.3$	$f=.1$	$f=.2$	$f=.3$
Mean	29.6	10.0	10.0	87.3	94.4	96.7
Trimmed mean	42.0	10.0	10.0	79.7	93.5	96.3
Median	43.4	10.0	10.0	40.4	93.3	96.3
Multi-Krum	52.9	42.7	16.9	86.4	92.4	97.2
CB-SAFE+	51.7	37.6	42.8	–	94.2	95.1

$f=0.1$ (≈ 3 dirty clusters, within trim capacity) the static rules work: trimmed mean 42.0%, median 43.4%, Multi-Krum 52.9%, versus 29.6% for the undefended mean and 59% clean baseline. At $f=0.2$ (≈ 5 dirty clusters) every coordinate-wise rule is pinned at exactly 10.0% (random guessing), not because the model diverges but because the laundered $\pm h$ population drives the aggregate to zero (Proposition 1), while Multi-Krum, which selects whole vectors rather than per-coordinate statistics, still reaches 42.7% (16.9% at $f=0.3$). Robust rules that rely on magnitude order statistics are structurally weakened by the very mechanism that provides privacy; vector-selection rules retain power.

CB-SAFE+ restores learning past the static breakdown (Table II). With loss-probe flags and temporal suspicion, the defense identified all 3 of 3 attackers at $f=0.1$ (51.7%, near the clean ceiling) and 8 of 9 sign-flip attackers with *zero* honest false positives at $f=0.3$, recovering to 42.8% accuracy where every static coordinate-wise rule sat at 10% and Multi-Krum at 16.9%, comparable to the 39–41% the median achieves *without any privacy* ($c=1$), i.e., the defense refunds the privacy dial’s cost in rounds rather than anonymity. At $f=0.2$ per-round flag filtering alone recovers 37.6% (exclusions are rare because $m=1$ laundered sums raise root loss only mildly); Multi-Krum’s 44% remains stronger in this moderate regime,

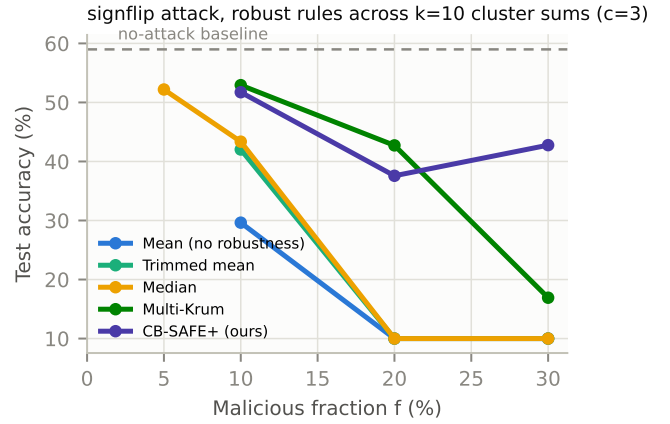


Fig. 3. Sign-flip attack ($\gamma=5$): accuracy vs. malicious fraction across aggregation rules over $k=10$ cluster sums ($c=3$). Coordinate-wise rules collapse at the contamination boundary the $(1-f)^c$ law predicts; vector selection degrades gracefully; CB-SAFE+ identifies and excludes the attackers.

so the defense’s advantage grows with attack severity and the two are complementary rather than ranked. Under label flipping (a weak attack in this regime: all rules land within 44–52%) the probe flags only the worst clusters and excludes conservatively (zero false positives at $f=0.2$), matching the best static rule.

Backdoor: a mechanism-independent blind spot. The trigger attack leaves main-task accuracy intact for every rule (50–55%) while ASR stays at 86–97% across mean, trimmed mean, Multi-Krum, and CB-SAFE+ (Fig. 4): backdoored cluster sums *reduce* clean root loss and align with the honest direction, so they are invisible to loss- and direction-based tests alike. The one partial exception is the coordinate-wise median at $f=0.1$, which suppresses ASR to 40%: order statistics can filter the trigger direction’s coordinates when contamination is sparse. We therefore claim CB-SAFE+ for *untargeted* model poisoning only; targeted trigger attacks under secure aggregation remain open, consistent with their difficulty even without privacy constraints.

E. Generalization Across Datasets and Modalities

To test whether the laundering effect and CB-SAFE+’s recovery are artifacts of one dataset, we repeat the sign-flip evaluation on FashionMNIST (a second image benchmark, three seeds) and on Edge-IIoTset (tabular IoT/IIoT intrusion detection with a multilayer perceptron, a different modality in the security domain of the closest prior work [7]). The pattern is identical everywhere (Table III, Fig. 5): every coordinate-wise rule collapses to chance at $f \geq 0.2$ (10.0% on the 10-class image sets, 7.3% on 15-class Edge-IIoTset), Multi-Krum degrades gracefully then falls at $f=0.3$, and CB-SAFE+ tracks the clean baseline. At $f=0.3$ CB-SAFE+ retains 84%, high-80s, and 97% of the no-attack accuracy on CIFAR-10, FashionMNIST, and Edge-IIoTset respectively, versus 12–18% for the worst static rule. Pooling all matched (dataset, f , seed) cells, CB-SAFE+ beats every baseline under a paired

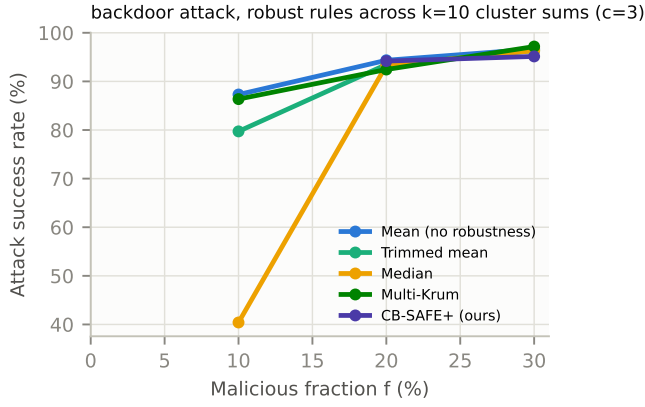


Fig. 4. Backdoor attack success rate vs. malicious fraction. No sum-level rule suppresses the trigger (median at $f=0.1$ partially excepted): backdoored sums help the main task and evade loss/direction probes, the honest scope limit of robustness under secure aggregation.

TABLE III
SIGN-FLIP ROBUSTNESS ACROSS DATASETS ($c=3$): FINAL TEST ACCURACY (%), MEAN OVER SEEDS. COORDINATE-WISE RULES COLLAPSE TO CHANCE AT $f \geq 0.2$ ON EVERY DATASET; CB-SAFE+ TRACKS THE CLEAN BASELINE. (CIFAR-10, FASHIONMNIST: 3 SEEDS; EDGE-IIoTSET: 1 SEED.)

Dataset	Rule	clean	$f=.1$	$f=.2$	$f=.3$
CIFAR-10 (images)	Median	59.0	47.8	10.0	10.0
	Multi-Krum	59.0	53.6	39.8	20.3
	CB-SAFE+	59.0	52.6	45.6	46.4
FashionMNIST (images)	Median	88.5	84.9	10.0	10.0
	Multi-Krum	88.5	85.9	83.0	43.1
	CB-SAFE+	88.5	85.8	86.1	84.1
Edge-IIoTset (tabular)	Median	63.7	59.6	13.6	7.3
	Multi-Krum	63.7	63.3	59.4	38.5
	CB-SAFE+	63.7	61.9	62.3	61.5

t -test with Holm correction: versus the mean by 48.3 points ($p < 10^{-4}$), the median by 37.7 ($p = 3 \times 10^{-4}$), and Multi-Krum by 11.9 ($p = 9 \times 10^{-3}$). The Edge-IIoTset result is the sharpest: at $f=0.3$ CB-SAFE+ holds 61.5% against a 63.7% clean baseline while mean, trimmed mean, and median all sit at 7.3%.

F. The Privacy–Robustness Dial, Measured

Fixing the rule (median) and sweeping cluster size traces the dial empirically (Fig. 6). Without privacy ($c=1$), the median degrades gently from 50% to 39% as f grows to 0.3, i.e., classical robust aggregation. At $c=3$ the same rule holds to $f=0.1$ (43.4%) and collapses beyond; at $c=5$ degradation begins already at $f=0.1$ (33.7%): the breakdown frontier shifts left exactly as $(1-f)^c$ predicts (Fig. 7). Two further observations sharpen the picture. First, at $f=0.05$ privacy is *free*: $c=3$ and $c=5$ match or slightly exceed $c=1$ (52.2% vs. 50.2%), because cluster averaging denoises before the median. Second, the dial’s cost is refundable: at $c=3$, $f=0.3$, where the private median collapses to 10% against 39–41% without privacy, CB-SAFE+ recovers comparable accuracy *while keeping the*

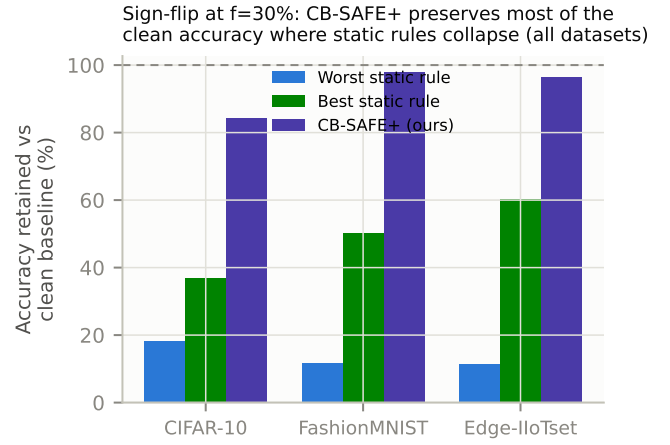


Fig. 5. Sign-flip at $f=30\%$: fraction of clean-baseline accuracy retained by the worst static rule, the best static rule, and CB-SAFE+, on every dataset. CB-SAFE+ preserves 84–97% where static rules keep 12–60%.

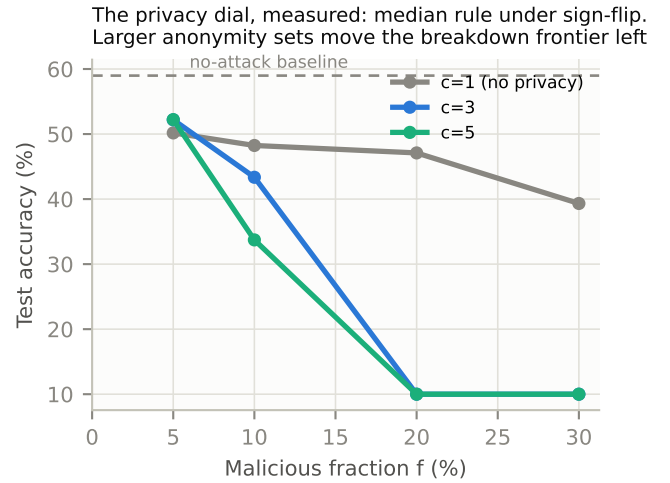


Fig. 6. The privacy–robustness dial, measured: median-rule accuracy under sign-flip for $c \in \{1, 3, 5\}$. Larger anonymity sets shift the breakdown frontier left, as $(1-f)^c$ predicts; at $f=5\%$ privacy costs nothing.

anonymity set, by paying in rounds (warmup) rather than in privacy.

VI. SECURITY DISCUSSION AND LIMITATIONS

What the design guarantees. Within a cluster, privacy reduces to the double-masking argument of [1] with the DH-style key agreement replaced by an IND-CCA2 KEM: masks are ChaCha20 outputs keyed by HKDF of encapsulated secrets, so an honest-but-curious server colluding with fewer than t cluster members learns only cluster sums. The robust layer never inspects anything finer than a cluster mean. **What it does not.** We do not claim security against an actively malicious server (who could misreport dropouts; standard mitigations apply), nor differential privacy on the cluster sums themselves; the anonymity set is bounded by the alive cluster size, and cluster sums of very small clusters leak more

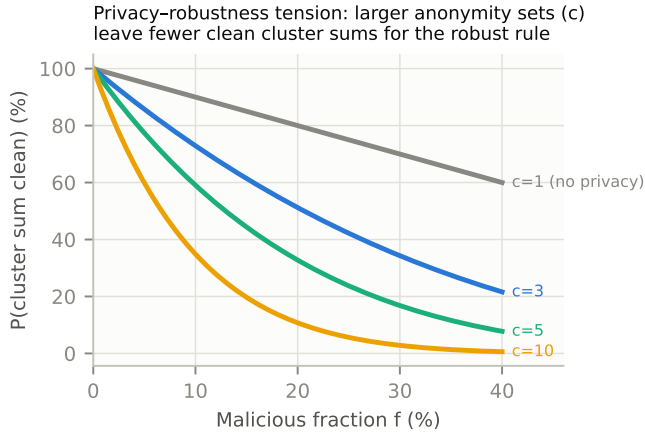


Fig. 7. The privacy-robustness dial, analytically: the anonymity set grows with cluster size c , while the probability that a cluster sum is uncontaminated decays as $(1 - f)^c$.

than full-cohort sums. Robustness degrades predictably as $k(1 - f)^c$ approaches the trim capacity: at $c=3$, $f=0.3$ leaves an expected 3.4 clean clusters of 10, near the breakdown of a 2-per-side trimmed mean; this is the price of the anonymity set and we report it rather than hide it. HQC operations are $\sim 200\times$ slower than ML-KEM's, which is immaterial here (setup-only) but would matter for protocols that encapsulate per round.

VII. CONCLUSION

CB-SAFE positions cryptographic diversity as a configuration choice rather than a research programme: the KEM sits behind one interface, and the measured cost of choosing the code-based occupant is a one-time setup constant. The more consequential finding is what secure aggregation does to robustness: cluster averaging does not merely dilute poisoning, it *launders* it, transforming conspicuous attacks into plausible-looking sums that defeat exactly the rules practitioners would reach for. Defending therefore requires signals that survive aggregation (direction, loss) accumulated over time. CB-SAFE+ shows one such design that identifies individual attackers from group observations without spending any privacy. The open questions our results make concrete: targeted backdoors remain invisible to sum-level probes at every malicious fraction we tested, an actively malicious server is outside our threat model, and the temporal identification argument should extend to an adaptive attacker who modulates its poisoning to stay below the probe margin; quantifying that game is the natural next step. Multi-seed replication and a second dataset are in progress; the implementation and all measurement scripts are released open-source as, to our knowledge, the first code-based reference point for post-quantum secure aggregation in FL.

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